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10 Low-resistivity nano-wires

Research areas: Chemistry, physics

Application: Nano-scale electronics

As wires shrink to nano scale, their resistivity grows, making them troublesome. Early in 2012, a multi-national team of researchers from USA and Australia demonstrated the possibility of fabricating low-resistivity nano-wires at extremely small scales by stringing together individual atoms in silicon. The wires, constructed from chains of phosphorus atoms, were just four atoms wide and one atom tall. Each wire was prepared by lithographically writing lines onto a silicon sample with microscopy techniques, then depositing phosphorus along that line. By packing the phosphorus atoms close together and encasing the nano-wires in silicon, the researchers were able to scale down without sacrificing conductivity—at least at low temperatures.

Volker Schmidt, a researcher at the Max Planck Institute of Microstructure Physics in Halle, Germany, commented in a press report that “Being able to fabricate metallic wires of such dimensions, by this theoretically microelectronics-compatible approach, could be a potentially interesting route for silicon-based electronics.”

The wires apparently have the carrying capacity of copper, so they are a good candidate for making chips. Considering their miniature size, they would help continue Moore’s law. However, the technique used by the researchers, namely, atom-scale lithography with a scanning tunnelling microscope, is still not industry-ready. So it might take many years before this becomes practicable.

“From now, over the next decade, scientists predict an explosion in scientific activity such as we have never witnessed before. Among others, one of the key technologies that would experience a dramatic growth would be computer power on the basis of breathtaking scientific advances... As transistor size keeps shrinking relentlessly, we can expect more and more powerful computers over the decades to come. We would eventually hit the wall of quantum physics laws that would possibly end silicon age and pave the way for totally new technologies that would have a profound impact on the way future computers would be architected and built. We would then be getting into largely unexplored and untested domains. These could be disruptive technologies such as optical, molecular, DNA and quantum computers. The microprocessors, as we know it today, would disappear. We must get ready for a completely synthetic human brain, moon mining, robotic moon bases, chips implanted in our brains, and what have you! Well, I believe that in the future a \$1,000 computer will have the processing power of the human brain,” signs off Venkat Ramana.

And, all these futuristic expectations will be fuelled by the scientific research of today! ●

The author is a technically-qualified freelance writer, editor and hands-on mom based in Chennai